Town of Quillfield Water Department

To	Select Board
From	Jaime Galloway, Water Superintendent, Town of Quillfield
Date	October 16, 2026
Re	Firm capacity on the Castlewick interconnect for 2027

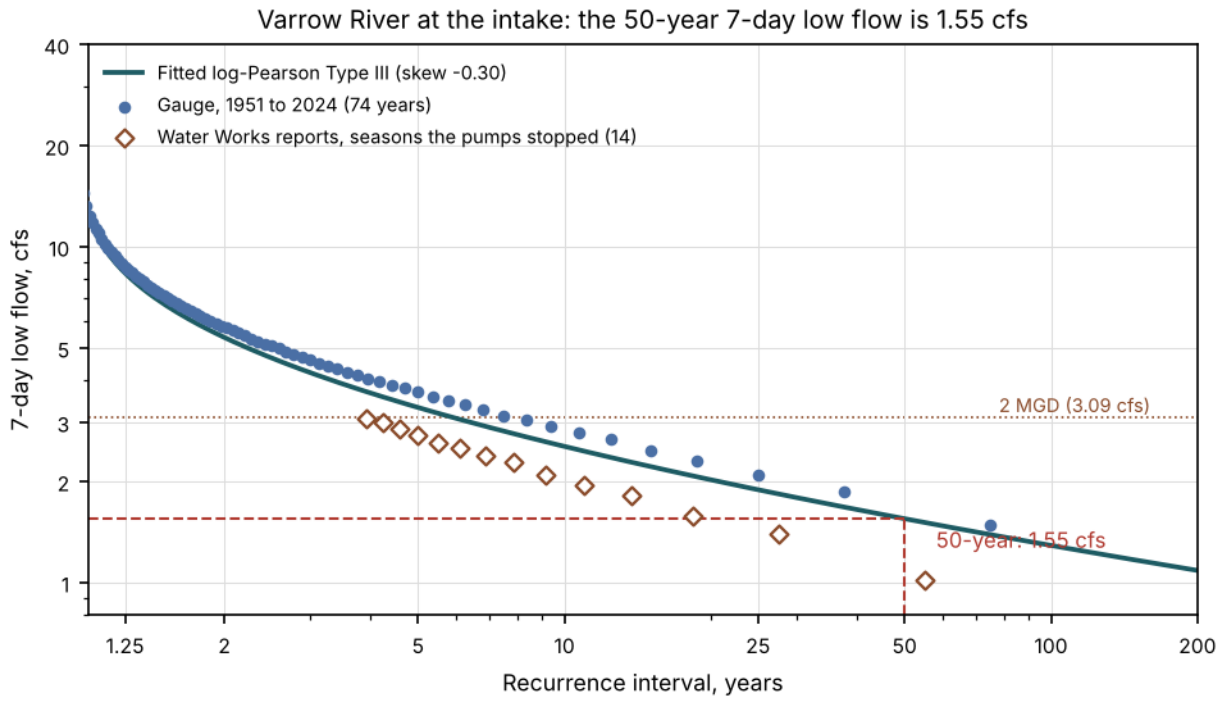
Reserve 0.3 million gallons a day (three blocks) of firm capacity on the Castlewick interconnect for 2027. At the river's 50-year 7-day low flow of 1.55 cubic feet per second our permit lets us take 0.68 MGD, 0.23 MGD short of the 2027 Stage 2 demand of 0.91 MGD; three blocks cover it, at a reservation charge of $85,200.

The design flow comes from Guidance WR-14's log-Pearson Type III, fitted by maximum likelihood to the whole record of the river at the dam: the Survey gauge's 74 approved climatic years from 1951 to 2024 and the Water Works reports from 1894 to 1950. The reports measured the river only in the 14 seasons the mill agreement stopped the pumps; in the other 40 reported years the pumps never stopped, so each of those years' 7-day low flow stayed above the agreement's two million gallons a day (3.09 cfs), and the fit takes them as that. The reports for 1909, 1918 and 1944 are missing and are left out.

Drought scenarios

Recurrence interval	7-day low flow, cfs	Withdrawal allowed, MGD	Stage 2 shortfall, MGD
10 years	2.54	1.32	none
25 years	1.89	0.90	0.01
50 years	1.55	0.68	0.23
100 years	1.29	0.51	0.40

Three blocks stop covering the Stage 2 shortfall in a drought rarer than the 65-year 7-day low flow (1.44 cfs).



Points plotted at Weibull positions, (n + 1)/rank, within the gauge's 74 years and within the 54 reported years. Climatic year 2025 is left out of the fit: its winter months are still provisional.

What our own system drew in past Stage 2 periods

Stage 2 in effect	Days	Average day, MGD	Highest day, MGD
2016-08-16 to 2016-10-02	48	0.81	0.90
2020-08-04 to 2020-08-26	23	0.86	0.93
2020-09-13 to 2020-10-04	22	0.78	0.86
2022-08-28 to 2022-09-14	18	0.85	0.90
2023-08-13 to 2023-09-29	48	0.84	0.90
2024-08-15 to 2024-09-22	39	0.85	0.93
2025-08-10 to 2025-09-17	39	0.87	0.93
2025-09-23 to 2025-10-11	19	0.79	0.84

Water supplied to the Quillfield distribution system: the plant's finished water less the Elm Hollow master meter, plus the Hollow Road wells (from the operator's round sheets since the wellfield telemetry failed on June 14, 2023). Flows in cubic feet per second, water in million gallons a day, both to two decimals; the shortfall and the block computed before rounding.